

An Exact MLP Encoding of Computation Graphs

This note constructs an *exact* realization of any computation graph as a ReLU MLP whose neuron groups are the graph’s nodes, and transfers the discrete complexity (D, N, r) to network depth and width.

1 Computation graphs and their complexity

Definition 1 (Computation graph). A *computation graph* is a DAG $G = (V, E)$ in which each node v carries a finite *alphabet* Σ_v (values $z_v \in \Sigma_v$) and a *parent set* $\text{pa}(v) = \{u : (u, v) \in E\}$, and each non-input node carries a *lookup table* $g_v : \prod_{u \in \text{pa}(v)} \Sigma_u \rightarrow \Sigma_v$, $z_v = g_v((z_u)_{u \in \text{pa}(v)})$. Vertices partition as $V = V_{\text{in}} \sqcup V_{\text{hid}} \sqcup V_{\text{out}}$; inputs have $\text{pa}(v) = \emptyset$ and are clamped, $z_v = x_v$ for $x \in \mathcal{X} := \prod_{v \in V_{\text{in}}} \Sigma_v$. Evaluating G computes z_v in topological order; the *function computed by G* is $f_G(x) = h((z_v)_{v \in V_{\text{out}}})$, with h the identity when $|V_{\text{out}}| = 1$.

We track four parameters, the last of which, Δ , governs network depth: the number of values $D := \max_v |\Sigma_v|$, the node count $N := |V|$ (with $N_{\text{hid}} := |V_{\text{hid}}|$), the max fan-in $r := \max_v |\text{pa}(v)|$, and the depth $\Delta := \max_v \text{dep}(v) \leq N$, where $\text{dep}(v) = 0$ on V_{in} and $\text{dep}(v) = 1 + \max_{u \in \text{pa}(v)} \text{dep}(u)$ otherwise. The quantity that transfers to the network is the total table content: node v has $\prod_{u \in \text{pa}(v)} |\Sigma_u| \leq D^r$ rows, each storing one symbol of Σ_v ($\log_2 |\Sigma_v| \leq \log_2 D$ bits), so the description length is

$$|G| := \sum_{v \notin V_{\text{in}}} \left(\prod_{u \in \text{pa}(v)} |\Sigma_u| \right) \log_2 |\Sigma_v| \leq N D^r \log_2 D. \quad (1)$$

We use a common alphabet size $|\Sigma_v| = D$ for readability; the per-node form replaces D^r by $\prod_{u \in \text{pa}(v)} |\Sigma_u|$ throughout, and no structural argument uses equality of the alphabets.

Running example G_0 . Figure 1(a) shows $\text{pa}(A) = \{x_1, x_3\}$, $\text{pa}(B) = \{x_1, x_2\}$, $\text{pa}(C) = \{B, x_3\}$, $\text{pa}(Y) = \{A, C\}$, $V_{\text{out}} = \{Y\}$, x_4 isolated: fan-in $r = 2$, depths 1, 1, 2, 3 so $\Delta = 3$, $N = 8$, and four D^2 -entry tables ($4D^2$ total). Nodes x_1 and x_3 are shared — x_1 feeds A, B and x_3 feeds A, C — exercising re-use rather than a disjoint partition.

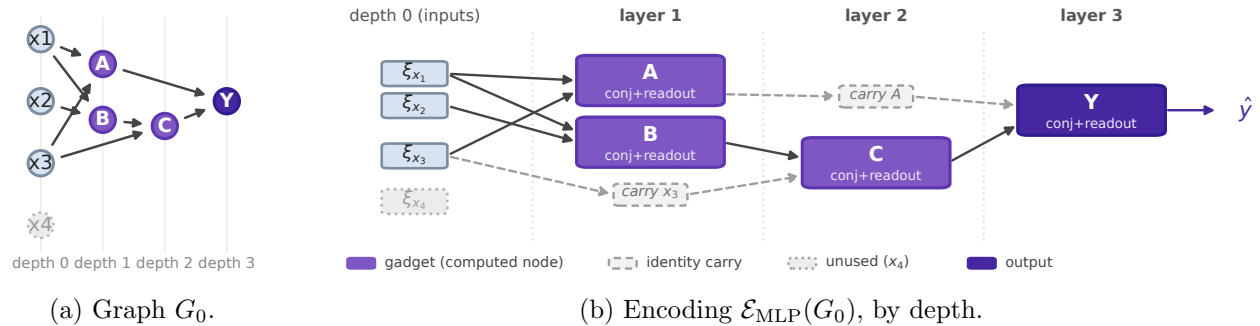


Figure 1: (a) The running example as a DAG. (b) Its exact MLP encoding by depth: computed nodes are conjunction–readout gadgets, deeper-consumed values are carried forward (dashed; the computed A via its one-time readout $\xi_A = U^A h^A$), x_4 is unused. Both panels produced by the figure script.

2 An exact MLP encoding

We encode each node value as a one-hot indicator. Write $\xi_v := e_{z_v} \in \{0, 1\}^{|\Sigma_v|}$ for the indicator of node v ; for inputs, $\xi_v = e_{x_v}$.

Per-node gadget. Fix a non-input v with parents $p_1, \dots, p_k$, $k \leq r$. A *conjunction layer* carries one unit per joint pattern $a \in \prod_i \Sigma_{p_i}$,

$$h_a^v = \text{ReLU}\left(\sum_{i=1}^k \xi_{p_i}[a_i] - (k-1)\right) = \mathbf{1}[z_{p_i} = a_i \ \forall i], \quad (2)$$

the argument being an integer equal to k iff every parent matches and $\leq k-1$ otherwise; so the vector h^v , with one component h_a^v per joint pattern a , is the one-hot indicator of the joint parent value. A *readout* $U^v \in \{0, 1\}^{|\Sigma_v| \times \prod_i |\Sigma_{p_i}|}$, $U_{c,a}^v = \mathbf{1}[g_v(a) = c]$, gives

$$\xi_v = U^v h^v = e_{g_v(z_{p_1}, \dots, z_{p_k})}. \quad (3)$$

The matrix U^v is the table; carrying no nonlinearity, it folds into the next stage's affine, so each graph depth costs one ReLU stage. Folding has a price: when a consumer with fan-in k reads a computed parent p (own fan-in r_p) from the preceding stage, the composed block carries exactly D^{r_p+k-1} nonzeros — for *every* table, since the block has at most D distinct rows whose supports $g_p^{-1}(c)$ partition the D^{r_p} columns ($\sum_c |g_p^{-1}(c)| = D^{r_p}$), each repeated over the D^{k-1} completions of the pattern. A node consumed more than one depth ahead is *carried* by the identity (ReLU fixes nonnegative one-hot entries); a *computed* one is first materialized once (U^v inside a later stage's affine, D^{r_v} nonzeros), then carried at cost D per stage. The prediction is $\hat{y} = \omega^\top \xi_Y$ (identity if the target is symbolic).

Proposition 1 (Exact realization and size). *Let G be a computation graph as above, with $W_{\max} := \max_d |\{v : \text{dep}(v) = d\}|$ the maximum number of nodes at one depth. The ReLU network $\mathcal{E}_{\text{MLP}}(G)$ satisfies $\mathcal{E}_{\text{MLP}}(G)(x) = f_G(x)$ for every $x \in \mathcal{X}$, using depth Δ (hidden ReLU stages), maximum width $m = O(W_{\max} D^r + ND)$, and $\sum_{v \notin V_{\text{in}}} \prod_{u \in \text{pa}(v)} |\Sigma_u| \leq ND^r$ conjunction units (plus $O(ND\Delta)$ carry units), with $O(NrD^{2r-1} + ND\Delta)$ nonzero weights in all (each folded parent slot costs exactly D^{r_p+k-1}).*

Proof. Induction on depth. At depth 0 the input groups hold $\xi_v = e_{x_v}$ by construction. Assume that at the input to stage d every node u with $\text{dep}(u) < d$ that is still needed has its indicator ξ_u available (by direct computation or carry). For each v with $\text{dep}(v) = d$ the parents have depth $< d$, so their indicators are present; (2) produces the joint indicator h^v and (3) produces ξ_v . Carries forward the remaining needed indicators by $\text{ReLU} \circ \text{Id}$. Thus the hypothesis holds at stage $d+1$. At $d = \Delta$ the output indicator ξ_Y is available and $\omega^\top \xi_Y$ returns the readout. The size statement counts the units in (2)–(3) per node and the carries; for the weights, each folded U^p contributes D^{r_p+k-1} per consumer slot ($\leq rD^{2r-1}$ per node), one-time materializations D^{r_v} (absorbed), and carries D per stage. $\square$

Remark 1. For worst-case tables the gadget storage $\Theta(ND^r)$ (weights) is order-optimal up to log factors, since an arbitrary table costs $\Theta(D^r \log D)$ bits per node by (1); structured g_v (linear, low-rank, monotone) compress below it. Across consecutive stages the forwarded activation has width $O(W_{\max} D^r + ND)$.

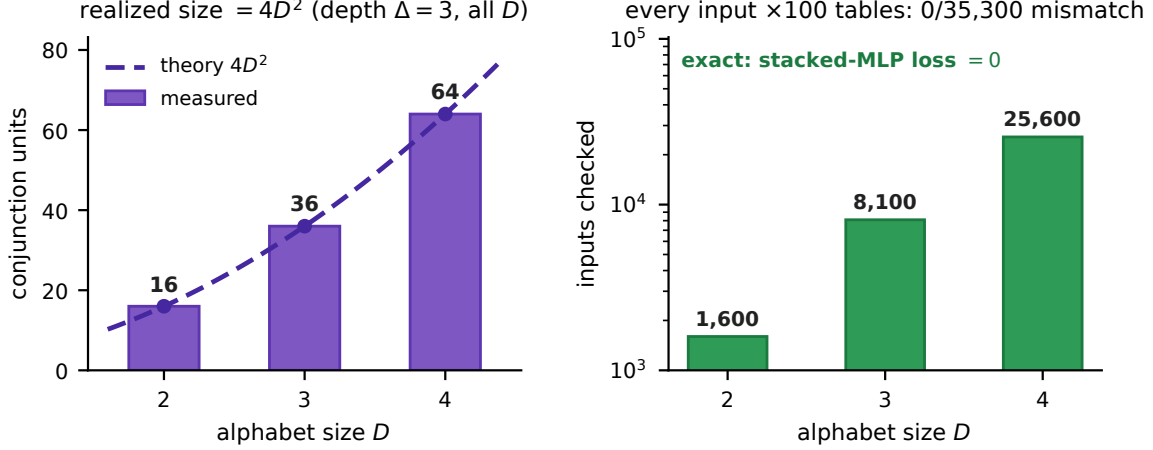


Figure 2: Empirical validation of $\mathcal{E}_{\text{MLP}}(G_0)$. *Left*: the realized conjunction-unit count equals the predicted $4D^2$, at constant depth $\Delta = 3$. *Right*: enumerating every input for $D \in \{2, 3, 4\}$ over 100 random table draws each (35,300 checks) gives *zero* mismatches, and a stacked ReLU MLP built from the per-depth weights attains exactly zero loss, its per-layer nonzero counts matching the predicted formulas on every draw. Produced by the validation script.

Complexity, and the dense comparison. Each conjunction unit is wired only to its $\leq r$ parents, and each readout — one nonzero per column of U^v standing alone — costs D^{r_v+k-1} per fan-in- k consumer slot once folded, so the gadgets carry $O(NrD^{2r-1})$ nonzero weights — the table content $|G|$ of (1) up to rD^{r-1} and $\log D$ factors — and the identity carries add $O(ND\Delta)$ (negligible once $\Delta = O(rD^{r-1})$), at depth Δ and width $m = O(W_{\max}D^r + ND)$. A *dense* ReLU MLP of the same depth and width is fully connected, with $\Theta(\Delta m^2)$ weights; the exact encoding is the sparse subnetwork whose wiring is G , using only an $O(NrD^{2r-1} + ND\Delta)/\Theta(\Delta m^2)$ fraction of them.

	depth	width	nonzero weights
$\mathcal{E}_{\text{MLP}}(G)$ (exact encoding)	Δ	$O(W_{\max}D^r + ND)$	$O(NrD^{2r-1} + ND\Delta)$
dense MLP, same depth & width	Δ	$O(W_{\max}D^r + ND)$	$\Theta(\Delta (W_{\max}D^r + ND)^2)$

Size and validation. For G_0 at $D = 4$: the four gadgets use $4D^2 = 64$ conjunction units and $2D^3 + 8D^2 + D = 260$ nonzero weights at depth $\Delta = 3$, the folded $B \rightarrow C$ and $C \rightarrow Y$ slots costing D^3 apiece (widest stage — A, B plus the x_3 carry — $2D^2 + D = 36$ units). A dense depth-3 MLP at the width bound $W_{\max}D^r + ND = 64$ holds $3 \cdot 64^2 \approx 1.2 \times 10^4$ weights — about $47\times$ more. An accompanying script reproduces the construction bit-exactly (Figure 2): enumerating all $D^{|V_{\text{in}}|}$ inputs for $D \in \{2, 3, 4\}$ across 100 random table draws each (35,300 checks) gives 0 mismatches, $4D^2$ conjunction units (16/36/64), $\Delta = 3$ ReLU stages, and per-layer nonzero weights ($4D^2 + D$, $D^3 + 2D^2$, $D^3 + D^2$) plus a D^2 head — matching the formula on *every* draw — and a stacked ReLU MLP assembled from the per-depth weights attains exactly zero loss.